SAMRATASHOKTECHNOLOGICALINSTITUTE
(Engineering College),VIDISHA M.P.
(An Autonomous Institute Affiliated to RGPV Bhopal)

Computer Science and Engineering

Semester/Year		I/II	Program		B.Tech.				
Subject Category	ESC	Subject Code:	CS26101	Subject Name:	**Fundamentals of Computer Science and Engineering**				
Maximum Marks Allotted						Contact Hours			Total Credits
Theory			Practical		Total Marks				
End Sem	Mid-Sem	Assignment/ Quiz	End Sem	Lab- Work		L	T	P	
70	20	10			100	3	0	0	3

Course Objective:

- To introduce the Computer Science and Engineering Concepts
- To understand the components of computers and generation, organization of computers.
- To familiarize students with the programming and problem-solving concepts using C Programming language.
- The course will help students to solve the problem using computer programming.

Course Outcomes:

Upon completion of this course ,the student will be able to:

- CO1:Familiarizetheimportanceofcomputerscienceandengineering.Understandthe concept of generation of computer and learn about component of computer system.
- CO2:UnderstandtheconceptofProblem-solvingusingCandImplementtheflowchartand program for solving Mathematical and Engineering problems.
- CO3:ArticulatetheModularProgrammingConcepttoResolvetheEngineeringProblems.
- CO4:ArticulatetheAdvanceCProgrammingConcepttoSolvetheEngineeringProblem using Structure, Union and File Management.
- CO5: Describe the basic concepts of pointers, file handling and their applications.

UNITS	Descriptions	Hrs.	CO's
I	**Introduction to Computer Science and Engineering**: Computer: Definition, Classification, Generation, Organization-Memory & Storage Systems. Introduction of Programming: characteristics, types, algorithm. Programming solving using computer concept, flowchart. Rules/conventions of coding, documentation, naming variables. Brief introduction to Linker, Loader, Assembler, Interpreter and Compiler	6	CO1
II	**Problem Solving using C**: History of C, Structure of a C Program; Data types, Constant & Variable, Operators - arithmetic, logical, bitwise, relational, ternary, expressions, Control Constructs – if-else, for, while, do-while, conditional looping, Switch-case statements, Escape statements, Special constructs – Break, continue, exit(), goto labels, Type conversion & type casting, Priority & associativity of operators; Type modifiers.	10	CO2
III	**Modular Programming:** Introduction to Arrays, Declaration and Initialization, Accessing Array Elements, Types of Arrays, Operations on Arrays, Arrays and Functions, Applications of Arrays.Functions; Arguments; Calling a function; Return statement; Parameter passing – call by value, call by reference; Scope, visibility and life-time rules for various types of variables; Storage classes.	9	CO3
IV	**Advance C Programming:** Structure – basic, declaration, membership operator, pointer to structure, referential operator, self-referential structures, structure within structure, array in structure, array of structures, type def. Union – basic, declaration; Recursion – basics, comparison with iteration, types of recursions- direct, indirect, tree and tail recursion, when to avoid recursion.	8	CO4
V	**Pointers:**Pointers, Pointers - & and * operators, pointer arithmetic, Arrays and Pointers, applications in memory management. **File Handling:** Basic file operations, including reading and writing data to files. Pre-processor Directives: C Pre-processor – basics, #Include, #define, Enumerated data type.	7	CO5
Total Hours		**40**	

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Subject Category	ESC	Subject Code:	**CS26101P**	Subject Name:	**Fundamentals of Computer Science and Engineering**					
Maximum Marks Allotted						Contact Hours			Total Credits
Theory			Practical		Total Marks				
End Sem	Mid-Sem	Assignment/ Quiz	End Sem	Lab- Work		L	T	P	
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List of Experiments

1. MakeaPosteronComponentofComputerSystems/GenerationofComputerSystemwiththeir working. (CO1)
2. Writeaprogram todeterminegivennumberisArmstrongnumberornot.(CO2)
3. Writeaprogram todeterminetherootsofquadraticequation $ax^2+bx+c=0$(CO2)
4. Writeaprogram tocalculatethefactorialofanintegerquantity.(CO2)
5. Writeaprogram toprintdiamondshapeusingstar.(CO2)
6. WriteaProgramtofind andprintthesumoffirstNPrimeNumbers.(CO2)
7. Writeaprogram toconvertbinarytodecimalanddecimaltobinary.(CO3)
8. Write a Program in C to read two arrays, add them and to print the resultant array. Use read_mat(),add_mat() and print_mat() functions. Array should not be declared as global variables. (CO3)
9. Write a program to read two matrix and apply addition, subtraction, multiplication,transpose operation and display result. (CO3)
10. WriteaCProgram tocalculateareaoftriangle,rectangle,circleusingfunction.(CO3)
11. Write a program using recursive function to output in reverse the sequence of characters input from the keyboard. The input is terminated by new line. Your output should be on a new line. Write an iterative solution for the same.
12. Write a Program to store data about 10 books. Which contain book title, price and number of copies of thebook.Afterreading thedataabout booksyourprogram shoulddisplaythe dataofallthebook which cost more than Rs 200. (CO4)
13. Writeaprogram using structure toaccept thecurrent timein(Hr:min:sec), update it byone secondand to print it. (CO4)
14. Write a program to count characters, spaces and new lines in a file. The name of the file should be entered through command line. (CO4)
15. Write the program using pointers to declare & initialize, accessing values, address operators, pointers size, passing by reference, array traversal, accessing array elements, pointers to an array, reverse an array string length and string copy.(CO5)

16. Write a C program to create a file and write contents, save, read the contents and close the file. (CO5)
17. Write a C program to count characters, words and lines in a text file.(CO5)

Text Book-

- LetusCByYashwantKanetkar,BPBPublication
- ProgramminginC,SchaumOutline,McGraw-Hill

Reference Books-

- ProgramminginANSI-CByE.Balagurusami,TMHPublication
- CProgramminglanguageByKernighan,Brian,W,Retchie,Dennis,PHIPublication
- InformationTechnology:TheoryandPracticeyPRADEEPK.SINHA(Author),PRITISINHA(Author)

Modes of Evaluation and Rubric

The evaluation modes consist of performance in Quiz/ Assignments, term work, and end-semester practical examinations.

List and Links of e-learning resources:

1. https://nptel.ac.in/courses/108/105/108105132/
2. https://de-iitr.vlabs.ac.in/

Recommendation by Board of studies on	June-2026
Approval by Academic council on	June-2026
Compiled and designed by	CSE & AI
Subject handled by department	CSE & AI

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3										2	3	1
CO2	3	2	3									2	3	1
CO3	2	3										2	3	1
CO4	2	3	2									2	2	1
CO5		3	2									1	3	1